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Resilient maintenance of carbonation-affected concrete infrastructure via physics-informed learning and predictive strategy

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ABSTRACT

Climate change accelerates carbonation in concrete, raising risks of cracking and spalling. However, existing model formulations often oversimplify this impact, inadequately representing the intrinsically non-linear and phase-dependent behavior of the carbonation process. In this paper, we propose a novel framework that integrates Physics-Informed Neural Networks (PINNs) and a Predictive Markov Decision Process with Phase-type Approximation (PMDP-PH), enabling resilient and cost-effective infrastructure maintenance under carbonation risks. PINNs embed governing physical laws within neural architectures, enabling accurate inference of carbonation dynamics even under limited observational data. This framework accommodates non-exponential sojourn time distributions in both the initiation and propagation phases, effectively approximated using hypo-exponential models. The PMDP-PH adaptively updates inspection and maintenance strategies by continuously refining the remaining useful life (RUL) distribution in real time. This decision-making process is formulated as a tractable multi-stage model predictive control (MPC) problem over selected belief states, ensuring a non-decreasing value function throughout the robust horizon. Applied to a representative infrastructure system, our method reduces total maintenance costs by up to 61.9% compared to benchmark strategies under variable deterioration scenarios. These findings highlight the promise of combining physics-informed learning with a new form of predictive control strategy to strengthen infrastructure resilience under the growing threat of carbonation-induced deterioration.

1. Introduction

Infrastructure components play a vital role in enhancing societal well-being, with the service life of these systems heavily dependent on the resilience of key materials, such as reinforced concrete [1]. As urbanization accelerates and the global population continues to grow, atmospheric concentrations of carbon dioxide are projected to rise significantly, leading to increased global temperatures. Climate change is expected to adversely affect the durability of concrete structures, potentially reducing the lifespan of existing infrastructure. For example, global warming could result in a 31% increase in structural damage to bridges, potentially shortening their service life by up to 15 years [2]. Consequently, the development of cost-effective maintenance strategies to mitigate the impacts of climate change is critical.

Carbonation is a major cause of deterioration in aging infrastructure, and its progression is further intensified by the effects of global warming. Existing studies commonly categorize the carbonation process into three distinct stages: initiation, propagation, and failure [3]. The initiation phase is marked by the ingress of carbon dioxide, which causes the de-passivation of reinforcement materials [4]. This phase is primarily governed by diffusion, with its rate significantly heightened by global warming and increased carbon emissions. However, pinpointing the exact end of the initiation phase is challenging due to limited inspection technologies, which can affect the evaluation of state resilience during the propagation phase and influence subsequent maintenance strategies. When the carbon dioxide concentration at the surface of the steel surpasses a critical threshold, the propagation phase commences [5]. In this phase, the rate of deterioration of the steel increases, potentially leading to a range of failure mechanisms such as cracking and spalling.

Climate change is accelerating the carbonation of concrete structures, thereby heightening the likelihood of cracking, spalling, and eventual structural failure. Current predictive models struggle to represent the inherently nonlinear and phase-dependent mechanisms driving carbonation. To our knowledge, research on developing adaptive and intelligent inspection and maintenance strategies specifically targeting this deterioran process remains limited. To bridge this gap, we

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Abbreviations

MDP	Markov Decision Process
SMDP	Semi-Markov Decision Process
PMDP-PH	Predictive Markov Decision Process with Phase-Type Approximation
POMDP	Partially Observable Markov Decision Process
POSMDP	Partially Observable Semi-Markov Decision Process
HMM	Hidden Markov Model
HSMM	Hidden Semi-Markov Model
PINN	Physics-Informed Neural Network
RUL	Remaining Useful Life
MPC	Model Predictive Control
PBVI	Point-Based Value Iteration algorithm
PPBVI	Predictive Point-Based Value Iteration algorithm

Notations

C	Concentration of carbon dioxide
D_s	Diffusion coefficient
v	Pore-water velocity
K_T	Rate constant for first-order decay
x	Diffusion depth
C_k	Initial carbon dioxide concentration within the concrete
C_s	Carbon dioxide concentration at the surface of the concrete
C_f	Carbon dioxide concentration at the interface between the concrete and steel
$X(t)$	Continuous-time Markov process
Q	Transition rate matrix
S_{ki}	Surrogated deterioration state
t_d	Predictive inspection timing
S_{t_d}	Underlying deterioration state at inspection timing t_d
O_{t_d}	Observation at inspection timing t_d
a_I	Inspection
a_1	No action
a_2	Preventive maintenance
a_3	Replacement
c_I	Cost of a_I
c_1	Cost of a_1
c_2	Cost of a_2
c_3	Cost of a_3
$P\left(S_{t_{d+1}}\|S_{t_d}, a_l\right)$	State–action transition probability
$E\left(O_{t_d}\|S_{t_d}\right)$	Emission probability
$\mathbf{B}\left(t_d\right)$	Belief state at inspection timing t_d
$F^{a_l}\left(t\|O_{0:t_d}\right)$	Estimated RUL distribution
$V\left(\mathbf{B}\left(t_d\right)\right)$	Value function of belief state $\mathbf{B}\left(t_d\right)$
R_T	Reliability threshold
$\mathcal{B}$	Belief set
$\mathcal{A}$	Action set
$\mathcal{O}$	Observation set
$\mathcal{S}$	State set
D_r	Robust decision epochs

introduce a novel methodological framework that combines Physics-Informed Neural Networks (PINNs) with a Predictive Markov Decision Process with Phase-type Approximation (PMDP-PH). This integration supports cost-efficient and resilient decision-making for infrastructure maintenance under carbonation-induced risks. Within the proposed approach, the carbonation process is recalibrated by embedding domain-specific physical laws into the PINN, while deterioration dynamics are represented as a surrogate partially observable Markov process through phase-type approximation. Inspection intervals are then adaptively optimized according to the predicted remaining useful life (RUL), with continuous updates informed by real-time inspection data. The key contributions of our proposed methodology are as follows:

1. *Recalibration of deterioration dynamics:* A novel combination of PINNs and phase-type (PH) distributions is developed to model and recalibrate infrastructure deterioration processes influenced by carbonation under climate change conditions.
2. *Proactive inspection and maintenance optimization:* A PMDP-PH is formulated, enabling predictive and proactive inspection and maintenance strategies. This approach achieves a 61.9% cost reduction compared to state-of-the-art methods.
3. *Predictive control with monotonic value:* The PMDP-PH is equivalent to a multi-stage model predictive control (MPC), offering additional properties and advantages, including contraction and yielding a non-decreasing value function.

The rest of this paper is structured as follows. Section 2 reviews the literature about the carbonation formulation and maintenance optimization. Section 3 presents the model formulation for carbonation. Section 4 optimizes inspection and maintenance decisions for carbonation. Section 5 presents an illustrative example to evaluate the effectiveness of the proposed approach. Section 6 concludes the paper.

2. Literature review

Carbonation corrosion is one of the common deterioration patterns of reinforced concrete. Contemporary research on deterioration mechanisms can generally be categorized into three major directions: physics-based, data-based, and hybrid approaches. In terms of the physics-based approach, the carbonation process is divided into three phases. The first stage is the diffusion of carbon dioxide, and the second stage is the corrosion of the steel until failure. Many scholars have conducted in-depth experimental studies on the factors that influence carbonation. Bao et al. conducted an experimental investigation together with multi-physics modeling of supercritical carbonation, aiming to examine how heterogeneously distributed coarse aggregates and material porosity influence the non-uniform penetration depth of carbonation in concrete [6]. Iwama and Maekawa developed a thermo-chemo-physical model across multiple scales to describe carbonation, decarbonation, and re-carbonation phenomena occurring during and following exposure to elevated temperatures [7]. The results from the physics-based approach are restricted to preset experimental conditions. In actual deterioration processes, environmental conditions fluctuate. The results based on physics models cannot be adapted to actual deterioration conditions. In terms of the data-based approaches, Duan and Cao introduced machine-learning-based models for carbonation and subsequently simplified them into a practical framework that demonstrated strong predictive accuracy on both accelerated and natural carbonation experiments [8]. Couto et al. employed Random Forest (RF), Support Vector Regression (SVR), and Artificial Neural Networks (ANNs) to estimate carbonation depth, utilizing a synthetic dataset of 20,000 samples derived from the validated Possan formulation [9]. Machine learning and deep learning can help predict the carbonation deterioration. However, in practice, high-quality data is hardly obtained, especially for the carbonation process. In addition, the lack of interpretability also limits the application of data-driven approaches.

As for the hybrid approach, Li et al. developed a hybrid surrogate approach, termed DL-LGBM-DRS, that merges deep learning with

the LightGBM algorithm to achieve reliable and efficient estimation of low-cycle fatigue life across a wide range of multiaxial loading cases [10]. PINNs provide a comprehensive framework for evaluating the sojourn time distribution. Tu et al. introduced a PINN that incorporates PointNet-based adaptive refinement to study crack propagation in industrial applications [11]. Song et al. introduced a reliability assessment method utilizing an Importance Sampling-enhanced Adaptive PINN, which was subsequently implemented in real-world engineering scenarios [12]. The hybrid methods can leverage the advantages of both physics-based and data-based approaches while mitigating the difficulty in data acquisition in practice, and they have the potential to improve carbonation modeling.

Climate change is accelerating carbonation in concrete infrastructure, increasing the risks of cracking, spalling, and structural failure, which requires effective maintenance strategies to mitigate the effect and avoid failure. In the extant literature, carbonation is often modeled as a Markov process [13] for optimizing maintenance. However, the real sojourn time of the carbonation condition is non-exponentially distributed in dynamic environments. Hence, the semi-Markov process [14] is proposed to model general distributed deterioration. Zambon et al. integrated a semi-Markov process with the carbonation mechanism to forecast deterioration [15]. Wang and Miao developed an optimal strategy for preventive maintenance by constructing a model based on the semi-Markov framework, specifically tailored for a balanced system [16]. Zheng et al. enhanced condition-based maintenance strategies by employing dynamic threshold adjustments within a system framework modeled using the proportional hazards approach, structured under a Semi-Markov Decision Process (SMDP) framework [17].

Visual inspection is commonly employed to detect deterioration caused by carbonation; however, it is often hindered by significant measurement uncertainty [18]. Consequently, the accuracy of the semi-Markov model in modeling deterioration is influenced. A solving approach is to incorporate inspection uncertainty into the model formulation, such as the Partially Observable Markov Decision Process (POMDP), Partially Observable Semi-Markov Decision Process (POSMDP), Hidden Markov Model (HMM), and Hidden Semi-Markov Model (HSMM). Zhang et al. presented an innovative framework for optimizing sequential inspection strategies in hierarchical multistate systems, where the decision-making process is modeled using a finite-horizon POMDP [19]. Cheng et al. represented a safety-critical system with partial observability using a hidden continuous-time Markov model, and proposed an integrated optimization approach that combines inspection scheduling with a condition-based mission abort strategy [20]. Guo and Liang introduced a predictive Markov decision process to enhance inspection and maintenance policies for multi-state systems with limited observability, achieved by integrating elements of both HMM and POMDP [21]. Arcieri et al. integrated Bayesian decision theory with POMDPs to develop a resilient maintenance planning approach that accounts for model uncertainty, specifically applied to railway system management [22]. Compare et al. designed a decision-support system grounded in POMDPs to guide operational planning in Gas Transmission Networks (GTNs). This framework incorporates degradation estimates generated by Prognostics and Health Management systems, while accounting for fluctuating demand and the impact of operational choices on the progression of system deterioration [23]. Khaleghei and Kim applied phase-type approximation to convert POSMDP into POMDP and identified structural properties to aid in solving [24]. Chen et al. designed an HMM incorporating autocorrelated observational data to effectively model the degradation processes in manufacturing systems [25]. Liu and Zhu introduced an innovative switching HSMM to depict the progression of equipment deterioration across different operational modes over time [26]. Eleftheroglou et al. introduced an advanced Similarity Learning HSMM for prognostics, designed to improve prediction accuracy, particularly when encountering anomalous patterns not represented in the training dataset [27].

The POMDP, POSMDP, HMM, and HSMM are often implemented under the scheme of periodic inspection. However, carbonation follows stochastic patterns. In the early stages, systems often operate normally and may not require immediate maintenance. Conducting inspections during this phase may not contribute to improving conditions. Conversely, as deterioration advances, it can escalate rapidly, potentially leading to unexpected failures between scheduled inspections. Hence, to address the risk of carbonation affected by climate change, the existing inspection and maintenance strategy requires to be readjusted. The development of intelligent and adaptive inspection and maintenance strategies offers significant potential for the future. Duan et al. proposed a maintenance approach that adapts to changing environmental conditions, utilizing revised conditional reliability estimates for mechanical systems [28]. Yang et al. introduced a comprehensive global-dynamic maintenance strategy tailored for systems composed of multiple deteriorating components. This approach enables immediate updating of the system's overall health status following the execution of each group maintenance operation [29].

In this paper, we aim to design a hybrid approach to accurately predict the deterioration trend and RUL, and to formulate intelligent and adaptive maintenance strategies for carbonation even in the face of climate change.

3. Model formulation for carbonation

To recalibrate the inspection and maintenance strategy for carbonation, we model the carbonation process as a surrogate partially observable Markov process. The carbonation process is divided into initiation and propagation phases. The evaluation of their sojourn time distributions is first performed using a PINN alongside a competing failure model. The resulting non-exponential sojourn time distributions are subsequently approximated by a phase-type distribution. This transformation enables the carbonation process to be represented as a surrogate partially observable Markov process, facilitating the incorporation of both stochastic deterioration and partial observability in the model. The formulation framework is demonstrated in Fig. 1.

3.1. Evaluation of sojourn time distributions for carbonation

The initiation phase is characterized by the infiltration of carbon dioxide. When the concentration on the steel surface reaches a critical value, this marks the onset of the propagation phase. The governing partial differential equation for a one-dimensional transport process involving carbon dioxide in the initiation phase can be expressed as [30]

$$\frac{\partial C}{\partial t} = D_s \frac{\partial^2 C}{\partial x^2} - v\frac{\partial C}{\partial x} - K_T C \tag{1}$$

where C represents the concentration of carbon dioxide (in g/m^3). x refers to diffusion depth (in m), and t is the time (in s). $\frac{\partial C}{\partial t}$ represents the change rate of the concentration of carbon dioxide over time. Eq. (1) incorporates the coupling of multiple mechanisms, including diffusion, convection, and reaction. D_s is the diffusion coefficient (in m^2/s). $\frac{\partial^2 C}{\partial x^2}$ denotes spatial diffusion of the concentration of carbon dioxide. v describes convective transport, where carbon dioxide is carried by the bulk motion of pore water (in m/s). $\frac{\partial C}{\partial x}$ expresses the concentration of carbon dioxide gradient along depth. K_T expresses the reaction rate constant for first-order decay at a specified temperature (in 1/s). The initial and boundary conditions are specified as $C(x,0) = C_k$, $C(0,t) = C_s$, and $C(L,t) = C_f$, where C_k, C_s and C_f correspond to the initial carbon dioxide concentration within the concrete, at the surface of the concrete structure, and at the interface between the concrete and steel, respectively. L represents the thickness of the concrete cover (in m).

A PINN is employed to assess the sojourn time distribution during the initiation phase. The architecture of the neural network consists

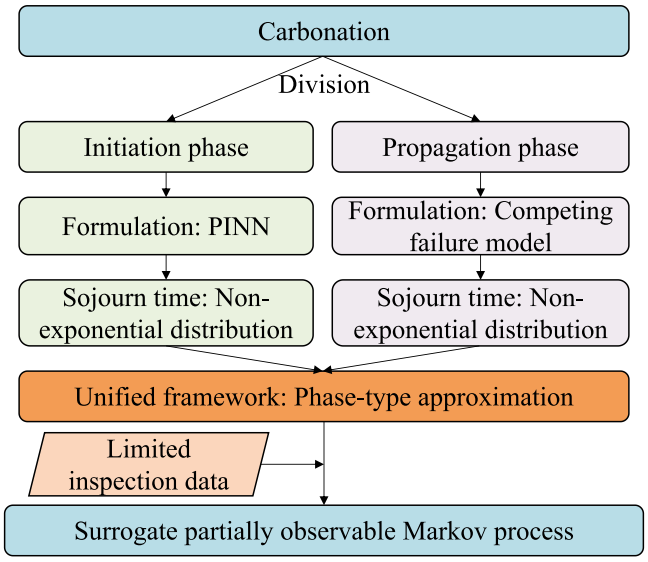


Fig. 1. The framework of model formulation for carbonation.

Table 1
The characteristics of the dataset.

Characteristic	Mean	Medium	Max	Min
Location (m)	10.5	10.5	20	1
Time (s)	0.03	0.03	0.05	0.01
Concentration (g/m^3)	0.6213	0.6300	0.6834	0.3699

of ten layers, as illustrated in Fig. 2. The first input layer contains 2 neurons, corresponding to x and t, while each of the eight hidden layers comprises 20 neurons. The output layer consists of a single neuron representing $C(x,t)$. The activation functions used in the hidden layers are Tanh. Learning rate schedule is 0.85. The optimization algorithm is L-BFGS. The convergence conditions include three criteria: (1) The maximum iteration is 50 000. (2) The L2-norm of the current gradient is less than 0.00001. (3) The change in the loss function value between two consecutive steps is smaller than 10^{-16}. To evaluate the performance of the network during training, we adopt the Mean Squared Error (MSE) as the loss function, formulated as shown in Eq. (2).

$$\mathcal{L} = \mathcal{L}_d + \mathcal{L}_{pde} + \mathcal{L}_{bcc} \tag{2}$$

The loss function incorporates the loss from the data, the partial differential equation, and the initial and boundary conditions as

$$\begin{aligned}
\mathcal{L}_d &= \frac{1}{n}\sum_{i=1}^{n}\left(C_i - C_{i,data}\right)^2 \\
\mathcal{L}_{pde} &= \frac{1}{n}\sum_{i=1}^{n}\left(\frac{\partial C_i}{\partial t} - D_s\frac{\partial^2 C_i}{\partial x^2} + v\frac{\partial C_i}{\partial x} + K_T C_i\right)^2 \\
\mathcal{L}_{bcc} &= \frac{1}{m_1}\sum_{i=1}^{m_1}\left(C_i(x,0) - C_k\right)^2 \\
&+ \frac{1}{m_2}\sum_{i=1}^{m_2}\left(C_i(0,t) - C_s\right)^2 + \frac{1}{m_3}\sum_{i=1}^{m_3}\left(C_i(L,t) - C_f\right)^2
\end{aligned} \tag{3}$$

where m_1, m_2, m_3 and n represent the sample sizes. C_i denotes the predicted value from the PINN. In our PINNs, we set the same weights for different losses.

We set $v = 10^{-12}$ m/s, $K_T = 10^{-10}$ 1/s, $D_s = 10^{-10}$ m^2/s, $L = 0.05$ m [30]. In the PINN, input parameters include location and time. The target variable is concentration. The dataset contains 100 samples. The characteristics of the dataset are listed in Table 1.

The concentration of carbon dioxide changes along with depth and time is demonstrated in Fig. 3(a). The initial and boundary conditions are incorporated to control the prediction of the PINN as shown with $\times$ in Fig. 3(a). Slice diagrams of concentrations for different depths are illustrated in Fig. 3(b) to reflect the accuracy of the PINN. To present the slice map at three distinct depths simultaneously, we scaled the coordinate axes accordingly. The relative error prediction (Euclidean norm) is 0.01793.

The significant variables include D_s diffusion coefficient, v pore-water velocity, and K_T rate constant for first-order decay at a specified temperature. D_s and K_T are affected by the temperature caused by climate change. Hence, we test the sensitivity of the two parameters by simulating the concentration change at a depth of 0.05 m. We uniformly sampled 20 values for one parameter within a specified range while holding the other parameter constant to conduct a sensitivity analysis. The results of the sensitivity test are demonstrated in Fig. 4. The results illustrate that D_s has more influence on the concentration change than K_T. Hence, we control the distribution of D_s to estimate the RUL in the initiation phase.

To describe the uncertainty of the initiation phase, we control the critical parameter D_s as Lognormal as $\mathcal{LN}(1^{-10}, 1^{-10}/8)$. We set the carbon dioxide concentration on the surface of the steel to reach 0.5 g/m^3 as the critical value. Then we obtain the sojourn time distribution of the initial phase with mean 5.36, standard deviation 3.55, and skewness 1.80. The coefficient of variation is computed as 0.66.

To assess stability and robustness, we perform k-fold cross-validation. We use the correlation coefficient (R), the mean absolute error (MAE), and the root mean squared error (RMSE) as metrics for comparing the Deep Neural Network (DNN). The results are demonstrated in Table 2. The PINN performs better when the sample size is small.

Based on our prediction of the initiation phase's end time, we can determine the onset of the propagation phase. During the propagation phase, reinforcement corrosion accelerates, leading to the development of critical failure modes, including cracking and spalling. These mechanisms contribute significantly to the deterioration of structural integrity. The deteriorations to crack and spalling are formulated as stochastic $Z_1(t)$ and $Z_2(t)$ for $t \geq 0$. The failure (crack or spalling) is self-revealing as denoted by F. We utilize a competing failure model to demonstrate this process and evaluate the sojourn time distribution. Here, we assume the failure of the crack and spalling is independent. Hence, the sojourn time of this propagation phase is

$$T_2 = \min\left\{t_1 : Z_1(t_1) = F; t_2 : Z_2(t_1) = F\right\} \tag{4}$$

To describe the uncertainty of the propagation phase, we set the distributions of t_1 and t_2 to follow $\mathcal{LN}(2.08, 0.33)$ and $\mathcal{LN}(2.79, 0.28)$

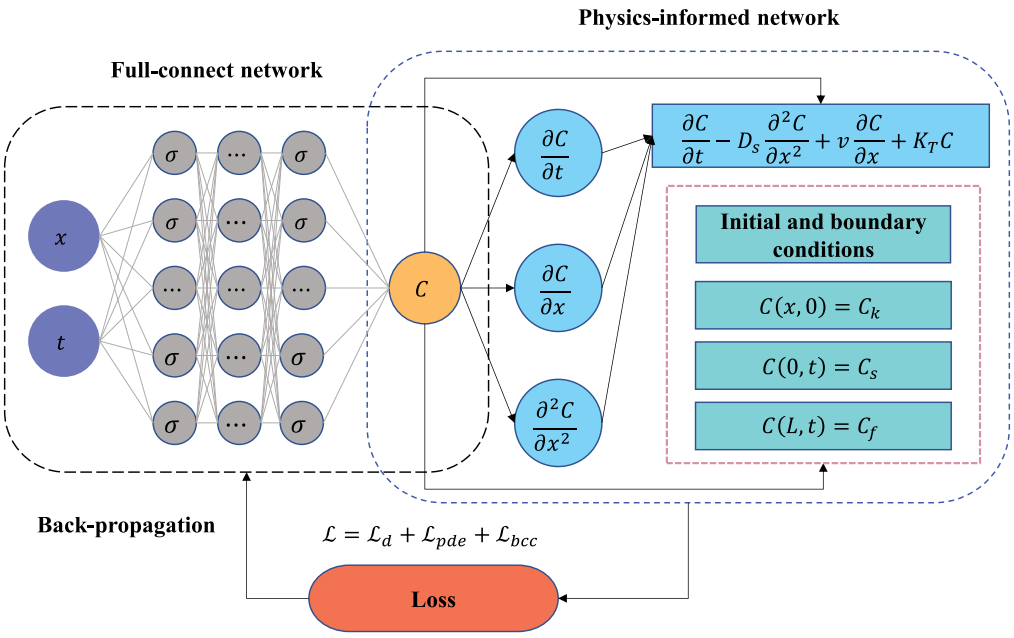
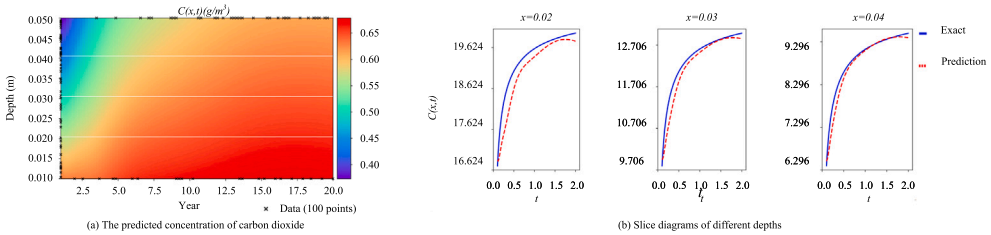


Fig. 2. The architecture of the PINNs.



(a) The predicted concentration of carbon dioxide (b) Slice diagrams of different depths

Fig. 3. The results of the PINNs.

Table 2
The comparative results between DNN and PINN.

Sample size	R		MAE		RMSE	
	DNN	PINN	DNN	PINN	DNN	PINN
40	0.918749	0.98999	0.006003	0.002819	0.01108	0.00414
60	0.84326	0.994047	0.007942	0.002411	0.020154	0.003031
80	0.99568	0.997537	0.003004	0.002056	0.004074	0.002885
100	0.996891	0.997883	0.003385	0.002769	0.004318	0.003732

according to [15]. The sojourn time distribution for the propagation phase is characterized by a mean of 8.34, a standard deviation of 2.66, and a skewness of 0.76. The coefficient of variation is calculated to be 0.32.

In our research, we utilize PINN to capture the deterioration of carbonation. More specifically, in the first stage of carbonation (Initiation phase), the carbon dioxide diffuses from the surface of the concrete until the concentration on the surface of the steel reaches a critical value. Therefore, we considered the changes in concentration values with both the depth of the concrete and the time dimensions. When simulating the initiation phase, we took into account the boundary conditions (i.e., the environmental settings). We used numerical values at discrete points to train the PINN network. The PINN enables continuous prediction of carbonation deterioration.

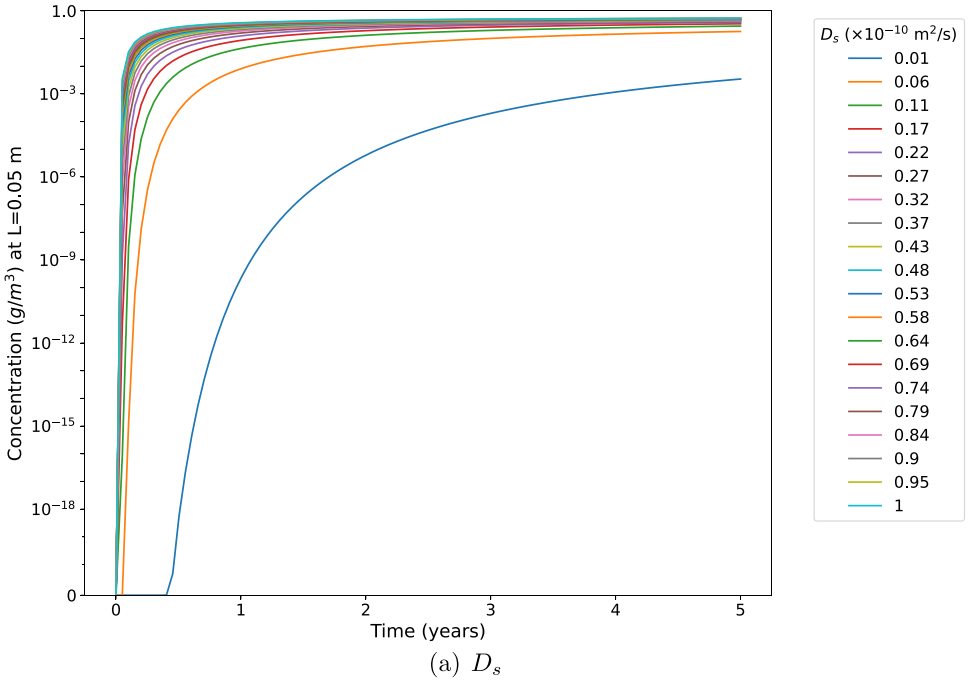
Through the PINN and competing failure models and controlling the variance of critical parameters, we simulate the uncertainty of the carbonation process. The estimated sojourn times of the carbonation phases follow non-exponential distributions. Hence, traditional Markov models are unsuitable for the formulation of carbonation. When the coefficient of variation is below 1, the distribution can be approximated by a hypo-exponential distribution [31,32], which is a specific form of phase-type distribution. This approximation allows the carbonation process to be effectively modeled using a surrogate Markov process.

3.2. Phase-type approximation

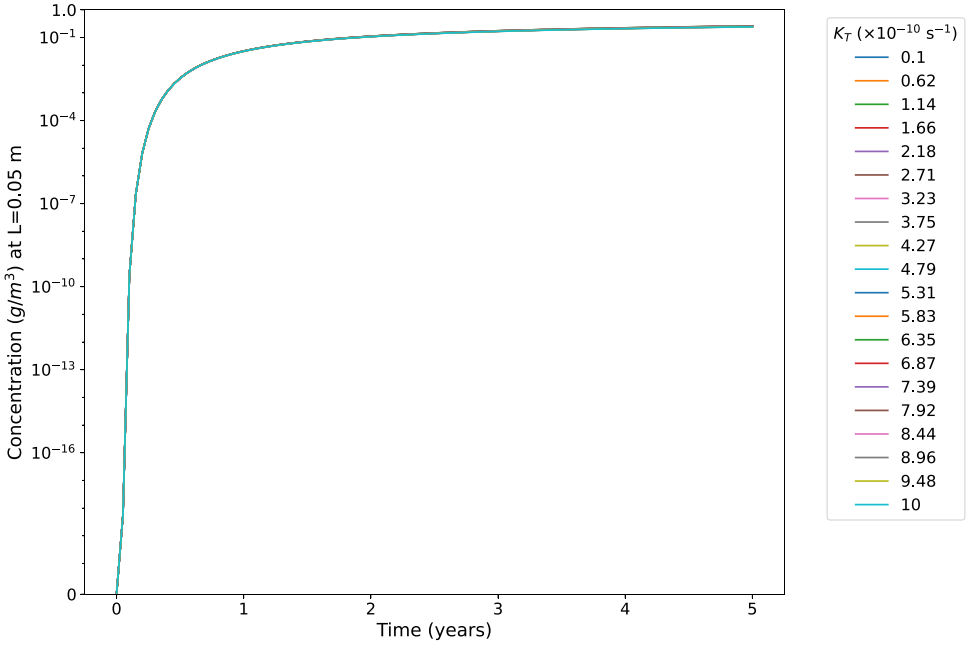
The hypo-exponential distribution describes the random time it takes for a Markov process to reach its absorbing state. Consider a continuous-time Markov process $\{X(t); t \geq 0\}$ with $n+1$ states, where the first n states are transient and state $n+1$ is absorbing. The initial state of the process is described by the probability vector $(\boldsymbol{\alpha}, a_0)$, where $\boldsymbol{\alpha}$ is a $1 \times n$ vector and a_0 is a scalar. The hypo-exponential distribution describes the time interval between the start of the process and its absorption into the absorbing state. The probability of transitioning between states within the time interval Δt is expressed as Eq. (5).

$$P\{X(t+\Delta t) = i+1 \mid X(t) = i\} = \lambda_i \Delta t + o(\Delta t) \tag{5}$$

where λ_i represents the transition rates from state $i = 1, 2, \ldots, n$ to the next state. Hence, the transition rate matrix Q follows the structure as



(a) D_s



(b) K_T

Fig. 4. The sensitivity test of D_s and K_T.

Eq. (6).

$$Q = \begin{bmatrix} -\lambda_1 & \lambda_1 & 0 & \vdots & 0 & 0 \\ 0 & -\lambda_2 & \lambda_2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \vdots & -\lambda_{n-1} & \lambda_{n-1} \\ 0 & 0 & 0 & \vdots & 0 & -\lambda_n \end{bmatrix} \tag{6}$$

The coefficient of variation (CV) of a hypo-exponential distribution is expressed as Eq. (7) [31,32].

$$CV = \frac{Var^2(t)}{E^2(t)} = \frac{E(t^2) - E^2(t)}{E^2(t)} \tag{7}$$

where $E(t) = \sum_{i=1}^{n} \frac{1}{\lambda_i}$, $E(t^2) = 2\sum_{i=1}^{n} \frac{1}{\lambda_i} \sum_{k=1}^{n} \frac{1}{\lambda_k}$. Hence, $CV < 1$.

Proposition 1 (hypo-exponential approximation). *The RUL distribution, approximated using a hypo-exponential model characterized by a probability vector* $\boldsymbol{\alpha}$ *and a transition rate matrix* Q*, can be represented as follows:*

$$F(t) = 1 - \boldsymbol{\alpha} e^{Qt} \mathbf{1} \tag{8}$$

Proposition 1 (hypo-exponential approximation) helps predict the RUL for a continuous-time Markov process. The entire corrosion process driven by carbonation can be divided into three phases: initiation, propagation, and eventual failure. As shown in Proposition 2 (hypo-exponential approximation), the hypo-exponential distribution preserves the closure property. Therefore, when this distribution is utilized for approximation, the overall deterioration process continues to conform to the principles of Markov modeling.

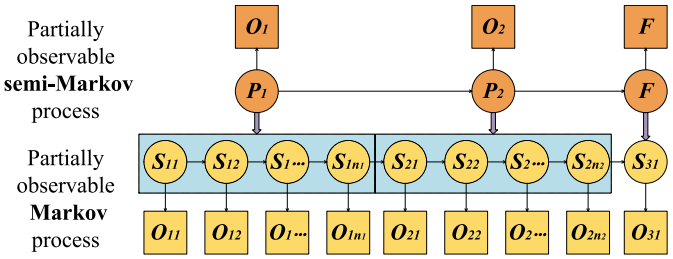


Fig. 5. Hypo-exponential approximation of carbonation.

Proposition 2 (hypo-exponential approximation). *Let PH_1 be a hypo-exponential distribution defined by the initial probability vector $\boldsymbol{\alpha}_1$, the transition rate matrix Q_1, and the failure rate vector $\boldsymbol{q}_1 = -Q_1\mathbf{1}$ and PH_2 be an independent hypo-exponential distribution characterized by $\boldsymbol{\alpha}_2$, Q_2, and $\boldsymbol{q}_2$. The convolution $PH_1 + PH_2$ then results in another hypo-exponential distribution with the following properties.*

$$\boldsymbol{\alpha} = \begin{pmatrix} \boldsymbol{\alpha}_1 & \vdots & 0 \end{pmatrix}. \tag{9}$$

$$Q = \begin{pmatrix} Q_1 & \vdots & \boldsymbol{q}_1\boldsymbol{\alpha}_2 \\ \cdots & \cdots & \cdots \\ 0 & \vdots & Q_2 \end{pmatrix}. \tag{10}$$

As indicated in Proposition 2 (hypo-exponential approximation), the sojourn time distribution, which is non-exponential in nature, can be effectively approximated using a hypo-exponential distribution. This allows the semi-Markov deterioration process to be reformulated as an equivalent Markov model, as illustrated in Fig. 5.

The original carbonation contains three phases: initiation phase (P_1), propagation phase (P_2), and failure (F). We utilize S_{11}, S_{12}, …, S_{1n_1} and S_{21}, S_{22}, …, S_{2n_2} to represent the surrogate states of P_1 and P_2 after hypo-exponential approximation. The state notation is defined as S_{ki}, where k expresses the phase and i represents the surrogate state. Failure indicates the end of carbonation and requires no hypo-exponential approximation. For uniform notations, the failure is expressed as S_{31} in the surrogate partially observable Markov process. The whole state set is defined as S. The observations of P_1 and P_2 are denoted as O_1 and O_2. The failure is self-revealing in our formulation. To match the surrogate Markov process, we need to discretize O_1 and O_2 into n_1 and n_2 observations based on the proportion of the mean sojourn time of the surrogate Markov states. For uniform denotation, the observation of failure F is defined as O_{31}. Hence, the discretized observations of the partially observable Markov process are denoted as $\mathcal{O} = \left\{O_{11}, \ldots, O_{1n_1}, O_{21}, \ldots, O_{2n_2}, O_{31}\right\}$. To apply the hypo-exponential approximation, the parameters (λ and n) are required to be assessed by moment matching. The mth moment of hypo-exponential distribution is expressed as

$$E\left(t^m\right) = (-1^m)\, m!\boldsymbol{\alpha}Q^{-m}\mathbf{1} \tag{11}$$

The parameters λ and n of the hypo-exponential distribution are determined by aligning its mean, standard deviation, and skewness with those of the target distribution. Hence, we set the goal of the objective function, as presented in Eq. (12), to reduce the deviation value in the mean, standard deviation, and skewness between the target distribution and its hypo-exponential approximation.

$$\arg\min_{\lambda_i, n} w_n = \left(\hat{\mu}_n - \mu\right)^2 + \left(\hat{\sigma}_n - \sigma\right)^2 + \left(S\hat{K}_n - SK\right)^2 \tag{12}$$

where $\hat{\mu}_n = \sum_{i=1}^{n} \lambda_i^{-1}$, $\hat{\sigma}_n = \sqrt{\sum_{i=1}^{n} \lambda_i^{-2}}$ and $S\hat{K}_n = \frac{2\sum_{i=1}^{n} \lambda_i^{-3}}{\left\{\sum_{i=1}^{n} \lambda_i^{-2}\right\}^{3/2}}$.

The hypo-exponential approximation will introduce an error in the model formulation. Here, we quantify the error as the deviation value w_n. We test the sensitivity of the hypo-exponential approximation with different combinations of surrogate states. The deviation values of the hypo-exponential approximation with different combinations of surrogate states are demonstrated in Fig. 6. The horizontal and vertical coordinates represent the number of surrogate states in the two phases. The deviation values are represented by a heat map. As the number of surrogate states increases, the deviation value shows a downward trend. Then we estimate the optimal maintenance strategies for different combinations of surrogate states. We consider three possible actions in the optimization of maintenance strategy: no action, preventive maintenance, and replacement, which are labeled as $\mathcal{A} = \{a_1, a_2, a_3\}$. No action does not influence the deterioration. Preventive maintenance is implemented to avoid failure, which restores the system to its initial state. The replacement is implemented only when an unexpected failure happens. The costs corresponding to these actions are represented as c_1, c_2, and c_3 respectively, where $c_1 = 0 < c_2 < c_3$, reflecting the typical increasing cost and downtime penalty associated with replacement. The optimal maintenance strategies for different combinations of surrogate states are demonstrated in Fig. 7. $P_1 : i$ and $P_2 : j$ represent that the P_1 and P_2 are approximated by i and j surrogate states. We use the horizontal axis to represent the cumulative duration of the surrogated states. Blue indicates no maintenance (a_1), yellow indicates preventive maintenance (a_2), and red indicates replacement (a_3). Fig. 7 illustrates that even though the hypo-exponential approximation introduces the deviation values, the optimal maintenance strategies remain consistent. In other words, the deviation value should be kept within a small range, which has no impact on the optimal maintenance strategies. We obtain the surrogate states and transition rates by hypo-exponential approximation in Table 3.

Through hypo-exponential approximation, we capture the carbonation with a surrogate partially observable Markov process. The synthesis of PINNs and phase-type distributions enables modeling and recalibrating infrastructure carbonation, even accelerated by climate change. Then we will present the optimization of the inspection and maintenance strategy.

4. Optimization of inspection and maintenance strategy

We propose a PMDP-PH approach to optimize inspection and maintenance strategies for mitigating carbonation effects. The proposed method updates inspection and maintenance strategies in real time, leveraging RUL estimations derived from partial observations. To further enhance the efficiency of the approach, we incorporate robust decision epochs to reduce computational complexity, thereby facilitating scalable and effective decision-making for strategies.

To cope with carbonation under climate change conditions, we reschedule the timings of inspections, represented as $\mathcal{T}_p = \{t_1, t_2, \ldots, t_p\}$. Due to the limitations of current inspection technologies, the deterioration state can only be partially observed. The relationship between actual deterioration and observable states is represented by the emission probability mapping, denoted as $E\left(O_{t_d} | S_{t_d} = S_{ki}\right)$. The state–action transition probability is expressed as $P\left(S_{t_{d+1}} = S_{k'i'} | S_{t_d} = S_{ki}, a_l\right)$. In contrast to the perfect observation scenario in a Markov Decision Process (MDP), the PMDP-PH utilizes belief states to capture our estimates of the deterioration, which are derived from inspection data. The belief is defined as $\mathbf{B}\left(S_{t_d} = S_{ki}\right) = P\left(S_{t_d} = S_{ki} | O_0, O_{t_1}, \ldots, O_{t_d}\right)$.

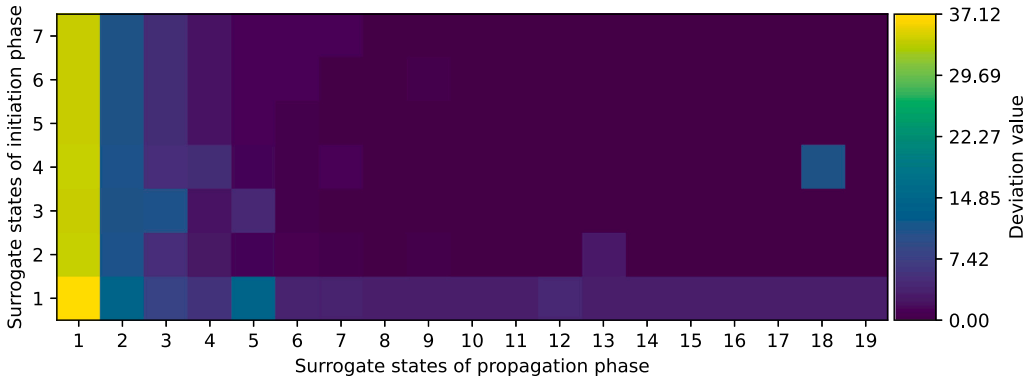


Fig. 6. The deviation values of the hypo-exponential approximation with different combinations of surrogate states.

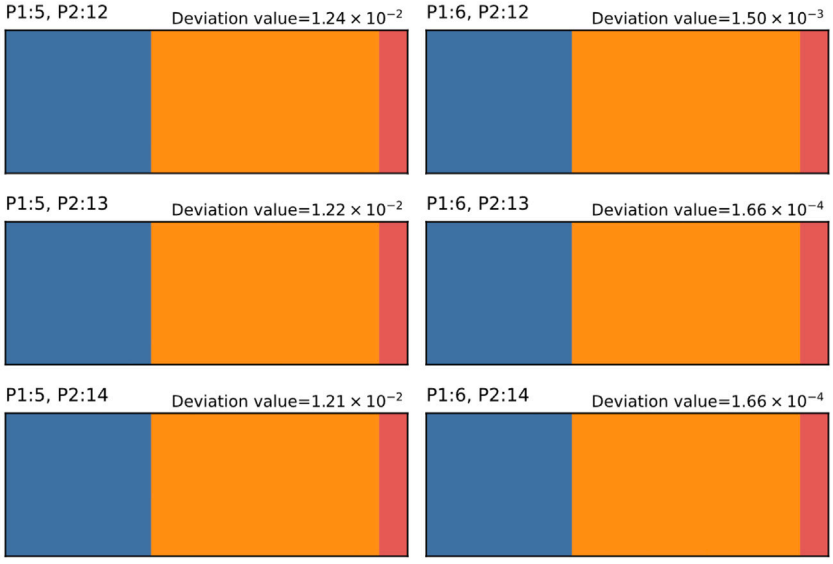


Fig. 7. The optimal maintenance strategies for different hypo-exponential approximations.

Table 3
The surrogate states and transition rates.

Phase	Number of surrogate state n	Transition rate λ_i
P_1	6	2.5831; 2.1497; 2.6594; 2.2975; 3.7618; 0.2915.
P_2	14	0.6619; 5.0342; 0.8348; 3.4873; 1.3594; 1.7567; 1.0300; 2.4669; 2.6575; 2.4588; 2.7645; 1.4456; 2.8966; 3.5321.

For simplicity, we utilize the notation $O_{0:t_d}$ to represent the sequence $O_0, O_{t_1}, \ldots, O_{t_d}$. The belief state $\mathbf{B}\left(t_d\right)$ consists of all $\mathbf{B}\left(S_{t_d} = S_{ki}\right)$ for each k and i at time t_d. Then the estimated RUL distribution is computed as

$$F^{a_l}\left(t|O_{0:t_d}\right) = \sum_{S_{ki}, S_{k'i'}} F_{S_{k'i'}}(t)\,\mathbf{B}\left(S_{t_d} = S_{ki}\right) P\left(S_{t_d^+} = S_{k'i'}|S_{t_d} = S_{ki}, a_l\right) \quad (13)$$

where $F_{S_{k'i'}}(t)$ indicates the estimated RUL distribution of the state $S_{k'i'}$. Its derivation can refer to Eq. (8). The timing for the next inspection is determined by ensuring that the reliability constraint R_T as

$$t^*_{d+1} - t^*_d = \left\{\tau^* : F^{a_l}\left(\tau^*|O_{0:t_d}\right) = 1 - R_T\right\} \quad (14)$$

Eq. (14) outlines an approach for scheduling future inspections based on the RUL prediction. Hence, the inspection strategy is adaptive to real deterioration. Maintenance optimization is conducted using the Bellman optimality equation. Unlike the MDP approach, the value function of PMDP-PH is defined over belief states. The value function is expressed as Eq. (15).

$$V\left(\mathbf{B}\left(t_d\right)\right) = \min_{a_l \in \mathcal{A}} \left[c_I + c_l + \gamma \sum_{S_{ki}} \mathbf{B}\left(S_{t_d} = S_{ki}\right) \sum_{S_{k'i'}} P\left(S_{t_d+\Delta} = S_{k'i'}|S_{t_d}, a_l\right) \sum_{O_{t_d+\Delta}} E\left(O_{t_d+\Delta}|S_{t_d+\Delta}\right) V\left(\mathbf{B}\left(t_d + \Delta\right)\right) \right] \quad (15)$$

where c_I denotes the cost of the inspection (a_I). The PMDP-PH is an online maintenance update model. Hence, we need to consider all possible observations after the time t_d with periodic decision interval

Δ. The convergence of Eq. (15) is demonstrated in Proposition 1 when $T \to \infty$. T denotes the horizon. We define an operation $\mathcal{H} : \mathbf{V} \to \mathbf{V}$. $\mathbf{V}$ denotes the vector of values for chosen belief states.

Proposition 1. The operation $\mathcal{H}$ is a contraction mapping on $\mathbf{V}$ in PMDP-PH.

The proof of Proposition 1 can be found in Appendix A. The belief is updated as Eq. (16).

$$\mathbf{B}\left(S_{t_d+\Delta}=S_{k'i'}\right) = \frac{\sum_{S_{ki}} \mathbf{B}\left(S_{t_d}=S_{ki}\right) P\left(S_{t_d+\Delta}=S_{k'i'}|S_{t_d},a_l\right) E\left(O_{t_d+\Delta}|S_{t_d+\Delta}\right)}{\sum_{S_{ki}}\sum_{S_{k'i'}} \mathbf{B}\left(S_{t_d}=S_{ki}\right) P\left(S_{t_d+\Delta}=S_{k'i'}|S_{t_d},a_l\right) E\left(O_{t_d+\Delta}|S_{t_d+\Delta}\right)} \tag{16}$$

Then Eq. (15) is transformed as Eq. (17).

$$V\left(\mathbf{B}\left(t_d\right)\right) = \min_{a_l\in\mathcal{A}}\left[\sum_{S_{ki}} \mathbf{B}\left(S_{t_d}=S_{ki}\right) q_{S_{ki}}^{a_l} + \gamma \sum_{S_{ki},S_{k'i'},O_{t_d+\Delta}} \mathbf{B}\left(S_{t_d}=S_{ki}\right) P\left(S_{t_d+\Delta}=S_{k'i'}|S_{t_d},a_l\right) E\left(O_{t_d+\Delta}|S_{t_d+\Delta}\right) V\left(\mathbf{B}\left(t_d+\Delta\right)\right)\right] \tag{17}$$

where $q_{S_{ki}}^{a_l} = c_I + c_l$. The initial condition is set as

$$V\left(\mathbf{B}\left(T\right)\right) = \sum_{S_{ki}} \mathbf{B}\left(S_T=S_{ki}\right) q_{S_{ki}}\left(T\right) = \mathbf{B}\left(T\right)\mathbf{q}\left(T\right) \tag{18}$$

where $q_{S_{ki}}(T)$ signifies the terminal value of the systems. Given that belief states are continuous, calculating the value function for these states can be computationally intensive. To simplify this process, we introduce Proposition 2, which addresses the value function.

Proposition 2. The optimized value of PMDP-PH is piecewise linear and convex expressed as

$$V\left(\mathbf{B}\left(t_d\right)\right) = \min_{\mathcal{L}}\left[\sum_{S_{ki}} \boldsymbol{\alpha}^{\mathcal{L}}\left(S_{t_d}=S_{ki}\right)\mathbf{B}\left(S_{t_d}=S_{ki}\right)\right], t_d < T \tag{19}$$

where $\mathcal{L}$ is a function about $O_{t_d+\Delta}$ and a_l. $\boldsymbol{\alpha}^{\mathcal{L}}\left(S_{t_d}=S_{ki}\right)$ is iteratively calculated as Eq. (20).

$$\boldsymbol{\alpha}^{\mathcal{L}}\left(S_{t_d}=S_{ki}\right) = q_{S_{ki}}^{a_l} + \gamma \sum_{S_{k'i'},O_{t_d+\Delta}} P\left(S_{t_d+\Delta}=S_{k'i'}|S_{t_d},a_l\right) E\left(O_{t_d+\Delta}|S_{t_d+\Delta}\right) \boldsymbol{\alpha}^{\mathcal{L}}\left(S_{t_d+\Delta}\right), t_d < T \tag{20}$$

The initial condition is set as $\boldsymbol{\alpha}^{\mathcal{L}}\left(S_T=S_{ki}\right) = q_{S_{ki}}(T)$.

The proof of Proposition 2 is provided in Appendix B. Proposition 2 shows that the process of iterating the value function can be reformulated as the determination of the $\boldsymbol{\alpha}$-vectors for the corresponding belief states. These $\boldsymbol{\alpha}$-vectors encode the optimal maintenance strategies. In the context of a continuous belief state space, the $\boldsymbol{\alpha}$-vectors partition the space into distinct regions, with each region associated with a specific optimal maintenance action. This approach simplifies the computation of the value function.

Proposition 3. The optimal $V\left(\mathbf{B}\left(t_d\right)\right)$ of the PMDP-PH is non-decreasing with respect to the stochastic increasing of $\mathbf{B}\left(t_d\right)$.

The proof of Proposition 3 is found in Appendix C. The definition of stochastic increasing is as follows: For vector $\mathbf{X} = \left[x_1, x_2, \ldots, x_n\right]$ and $\mathbf{Y} = \left[y_1, y_2, \ldots, y_n\right]$, $\mathbf{X}$ is said to be stochastically smaller than $\mathbf{Y}$, denoted by $\mathbf{X} \leq_{SI} \mathbf{Y}$ if the condition $\sum_{i=k}^{n} x_i \leq \sum_{i=k}^{n} y_i$ for any $0 \leq k \leq n$ is satisfied. Proposition 3 indicates that the optimal strategy in the PMDP-PH is a form of control-limit strategy.

Traditional value iteration and policy iteration algorithms are not practical for partially observable Markov deteriorating processes because of the significant computational complexity arising from the continuous belief state space. To address this, we propose a Predictive Point-Based Value Iteration (PPBVI) algorithm, which extends the Point-Based Value Iteration (PBVI) method by incorporating adaptive inspections. We focus on specific belief states along the deterioration trajectory. The representativeness of the selected belief states is measured by the gap, defined as $gap = \max\left(V\left(\widetilde{\mathcal{B}}\right) - V\left(\mathcal{B}\right)\right)$. $\widetilde{\mathcal{B}}$ represents the chosen belief states, and $\mathcal{B}$ denotes the full continuous belief states. As demonstrated by Proposition 2, the continuous belief state space can be partitioned into multiple regions. The value function can then be reformulated as Eq. (21).

$$V\left(\mathbf{B}\left(t_d\right)\right) = \min_{a_l\in\mathcal{A}}\left[\sum_{S_{ki}} \mathbf{B}\left(S_{t_d}=S_{ki}\right) q_{S_{ki}}^{a_l} + \min_{\boldsymbol{\alpha}^{\mathcal{L}}} \gamma \sum_{S_{ki}} \mathbf{B}\left(S_{t_d}=S_{ki}\right) \sum_{S_{k'i'},O_{t_d+\Delta}} P\left(S_{t_d+\Delta}=S_{k'i'}|S_{t_d},a_l\right) E\left(O_{t_d+\Delta}|S_{t_d+\Delta}\right) \boldsymbol{\alpha}^{\mathcal{L}}\left(S_{t_d+\Delta}\right)\right] \tag{21}$$

By leveraging specific belief states together with $\boldsymbol{\alpha}^{\mathcal{L}}$-vector data, the optimal strategy can be determined. Next, we examine the computational complexity involved. At each decision point, each $\boldsymbol{\alpha}^{\mathcal{L}}$ vector produces $|\mathcal{A}| \times |\mathcal{O}|$ new vectors for the next decision epoch, corresponding to all possible $\boldsymbol{\alpha}^{\mathcal{L}}$. In the algorithm, unnecessary $\boldsymbol{\alpha}^{\mathcal{L}}$ vectors can be pruned to mitigate computational demands. Importantly, the computation of $\boldsymbol{\alpha}^{\mathcal{L}}$ is independent of $\mathcal{B}$, allowing the results to be cached and reused for different belief states in $\mathcal{B}$. The computation of each $\boldsymbol{\alpha}^{\mathcal{L}}$ necessitates $O(|S|)$ operations to compute inner products, as described in Eq. (19). The overall complexity for calculating the value function, as indicated by Eq. (21), is $O(|\mathcal{A}| \times |\mathcal{O}| \times |\mathcal{B}| \times |S|^2)$. When computing the optimal strategy for all $\mathcal{B}$ using $\boldsymbol{\alpha}^{\mathcal{L}}$, the total number of operations required is $O(|\mathcal{A}|\times|\mathcal{O}|\times|\mathcal{B}|^2\times|S|^2)$. The whole computation complexity in the finite horizon is $O(|\mathcal{A}|\times|\mathcal{O}|\times|\mathcal{B}|^2\times|S|^2\times|T|)$. In other words, as the number of decision epochs increases, the computational complexity grows rapidly. To mitigate the computation, we implement robust decision epochs to optimize inspection and maintenance online. The optimization of the PMDP-PH can be regarded as a special case of multi-stage MPC [33] in Proposition 4.

Proposition 4. Given $\mathcal{S}$, $\mathcal{A}$, and $\mathcal{O}$, the optimization of the PMDP-PH with chosen belief states can be translated as a problem of multi-stage MPC.

The proof of Proposition 4 is found in Appendix D. Proposition 4 provides a link between the PMDP-PH model and the multi-stage MPC. The PMDP-PH model can be regarded as a special multi-stage MPC with contraction mapping and a non-decreasing value function. We assume that the uncertainty arising from deterioration, observations, and actions remains unchanged after a certain number of decision epochs, referred to as robust decision epochs, and persists until the end of the finite horizon. Hence, the computation complexity is reduced to $O(|\mathcal{A}|\times|\mathcal{O}|\times|\mathcal{B}|^2\times|S|^2\times|D_r|)$, which is smaller than $O(|\mathcal{A}|\times|\mathcal{O}|\times|\mathcal{B}|^2\times|S|^2\times|T|)$. Because robust decision epochs are smaller than the finite horizon $D_r < T$. The pseudocode for the PPBVI algorithm is provided in Algorithm 1.

In the PPBVI algorithm, a set of representative belief states is chosen to compute the value function. These belief states are derived from the uncertainty present in the multi-stage MPC models, based on discrete scenarios. Our approach is an iterative optimization process. The optimization is divided into two steps. The first step is to optimize the maintenance at the current inspection timing. In the second step, the next inspection timing is determined based on the reliability threshold and optimized action. Repeating the above two steps, we obtain the optimal inspection timings and maintenance strategy in the finite horizon.

Algorithm 1 PPBVI algorithm

Input: costs c_I, $c_1 = 0$, c_2, c_3, initial value $\mathbf{q}(T)$, initial belief state $\mathbf{B}(0)$, state-action transition probabilities $P\left(S_{t_d+\Delta} = S_{k'i'} | S_{t_d} = S_{ki}, a_l\right)$ for $\forall\ k, k', i, i', a_l$, emissions $E\left(O_{t_d+\Delta} | S_{t_d+\Delta} = S_{k'i'}\right)$, $\forall\ k', i', O_{t_d+\Delta}$, and RUL distribution $F\left(S_{k'i'}\right)$ $\forall k', i$.

Output: adaptive inspection timings $t_1^*, t_2^*, \ldots, t_p^*$ and maintenance strategy π^*

Initialization: Δ, $r\Delta$ (robust decision epochs)
for $t_d = 0, \ldots, T$ **do**
  for $t = t_d + r\Delta \le T, \ldots, t_d$ **do**
    calculate $\boldsymbol{\alpha}^{\mathcal{L}}\left(S_t = S_{ki}\right)$ based on Eq. (20)
    prune unnecessary $\boldsymbol{\alpha}^{\mathcal{L}}$
    calculate $V(\mathbf{B}(t))$ based on Eq. (21)
  end for
  obtain maintenance strategy $\pi\left(t_d\right)$
  //optimization of maintenance
  optimize t_{d+1} based on Eq. (14)
  //schedule of inspection timings
end for

5. Case study

To model carbonation in concrete infrastructure, we employ a surrogate partially observable Markov process to incorporate the characteristics of stochastic deterioration and partial observation. We propose a PMDP-PH to enable a shift from traditional condition-based maintenance to more predictive and adaptive maintenance practices tailored for carbonation.

5.1. Comparison between the PMDP-PH and the POMDP

In this study, we compare the performance of the PMDP-PH model with that of the POMDP. The POMDP is the state-of-the-art condition-based maintenance approach under a periodic inspection scheme with the assumption of partial observation and stochastic deterioration. Both models share identical assumptions and parameters. The transition probabilities among the interval $t_{d+1} - t_d$ between state S_{ki} and $S_{k'i'}$ can be computed as follows.

$$P_{S_{ki}S_{k'i'}}\left(t_{d+1} - t_d\right) = e^{Q*\left(t_{d+1} - t_d\right)}_{S_{ki}S_{k'i'}} \tag{22}$$

The elements of Q are demonstrated in Table 3. The emission is often assumed to follow the Gaussian distribution [24]. The probability of inspecting the actual state is 0.9. The remaining probabilities are evenly distributed among adjacent states. The cost of preventive maintenance and replacement is set as \$1500 and \$25 000. To assess the performance under varying deterioration conditions, we evaluate five distinct scenarios, as demonstrated in Fig. 8. According to the hypo-exponential approximation, carbonation is transformed by a Markov model of 21 surrogate states. The horizontal axis in Fig. 8 represents the state, while the vertical axis represents the time. In five scenarios, both PMDP-PH and POMDP can evaluate and update the estimation of the underlying state based on observations. We use a heat map to represent the change in belief probability. The optimal strategies are displayed on the right side of the subfigures. The POMDP relies on periodic inspections. The PMDP-PH schedules non-periodic inspections based on the estimated RUL; both approaches satisfy the reliability requirement and enable timely maintenance to mitigate risk. Specifically, PMDP-PH can reduce unnecessary inspections compared with the POMDP.

When the initial system state is good, the PMDP-PH achieves greater cost savings, as shown in Table 4. The PMDP-PH model demonstrates superior performance compared to the POMDP, achieving an average cost reduction of 61.9%, independent of the initial conditions. The cost savings are mainly derived from the reduction of non-essential inspections. In the PMDP-PH model, inspection schedules are predicted based on the estimated distribution of RUL. Each observation provides valuable insights into the system's deterioration state, allowing for adjustments to the inspection strategy in response to its stochastic nature. This approach, optimized through the use of inspection data, informs future inspection planning and maintenance decisions, ultimately improving cost-effectiveness regardless of initial system conditions.

Table 4
The cost comparison between PMDP-PH and POMDP.

Initial state	Cost(\$)		Reduction
	PMDP-PH	POMDP	
1	524.52	2534.086	79.3%
3	697.1432	2693.313	74.1%
5	1265.253	2835.347	55.4%
7	1989.198	3998.764	50.3%
9	1989.198	3998.764	50.3%
Average			61.9%

Table 5
The comparison of the computation time between PMDP-PH and POMDP.

Initial state	Time (s)		Reduction
	PMDP-PH	POMDP	
1	0.007	0.018	61.1%
3	0.012	0.029	28.6%
5	0.011	0.027	59.3%
7	0.011	0.026	57.7%
9	0.011	0.027	59.3%
Average			59.2%

Table 6
The comparison between PMDP-PH and POMDP under different cost settings.

Cost(\$)	Cost(\$)		Reduction
	PMDP-PH	POMDP	
500	1699.23	2155.02	21.5%
800	2204.81	2712.88	18.7%
1000	2541.86	3084.78	17.6%
1200	2878.92	3456.69	16.7%
1500	3384.5	4014.54	15.7%

The computation times between the PMDP-PH and POMDP are listed in Table 5. Among different scenarios, the PMDP-PH performs better than the POMDP with an average reduction in the computation time of 59.2%. Because of the robust horizon, the iterative number of PMDP-PH is reduced, which contributes to the reduction of computation time.

5.2. Sensitivity analysis

In practice, the real deterioration state is partially observable due to the limitations of current inspection technologies. Often, the emission probability is assumed to be a Gaussian distribution. Accordingly, we adopt this assumption and analyze the sensitivity to the emission probability. We set the probability of inspecting the actual state to 0.9, 0.7, and 0.5. The remaining probabilities are evenly distributed among adjacent states. The belief evolution and optimal maintenance strategy are demonstrated in Fig. 9.

Observation accuracy affects the estimation of the true deterioration state and the resulting maintenance strategy. Higher monitoring accuracy enables more accurate estimation of the true deterioration state, leading to more precise maintenance strategies.

We compare PMDP-PH and POMDP for preventive-maintenance costs ranging from \$500 to \$1500. The results are demonstrated in Table 6. Across cost settings, our approach reduces costs primarily by avoiding unnecessary inspections.



Fig. 8. The comparison of optimal maintenance strategies between the PMDP-PH and POMDP.

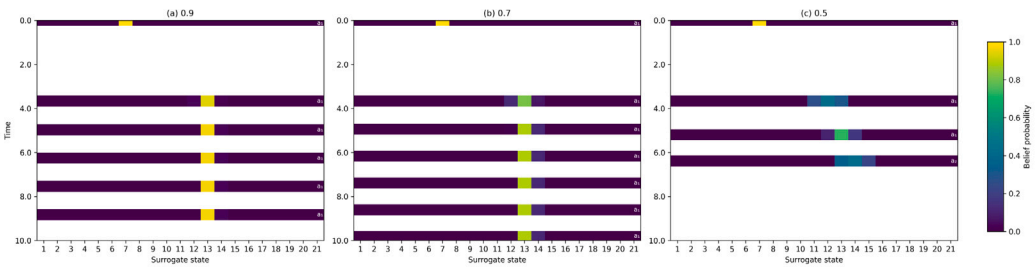


Fig. 9. The belief evolution and optimal maintenance strategy when the emission probability changes.

5.3. Discussion

In recent years, research efforts have increasingly examined how to optimize inspection and maintenance planning using POMDP models under periodic inspection settings. The resulting decision framework is generally categorized as condition-based maintenance. Unlike the POMDP formulation, the PMDP-PH requires inspection intervals to be forecasted and optimized through RUL estimation, thereby positioning it as a predictive maintenance strategy. Our proposed method adaptively determines future inspection moments in real time according to the estimated RUL. Relative to conventional periodic inspection schemes, this method is capable of reducing redundant inspections. Based on the above discussion, three essential remarks can be highlighted:

1. As demonstrated in Fig. 8 and Table 4, the PMDP-PH framework adaptively modifies inspection intervals by incorporating the latest inspection information into its RUL estimates. This dynamic strategy contributes to a reduction in superfluous inspections.
2. Table 5 highlights that the PMDP-PH has more computational efficiency than the POMDP when the robust horizon is applied.
3. Fig. 9 and Table 6 illustrate that the PMDP-PH enables optimizing the maintenance strategies and reducing essential inspections under different emission and cost settings.

The rationale for cost reduction arises from the PMDP-PH model's ability to adjust inspection intervals according to outcomes of prior assessments, thereby minimizing unnecessary inspections over the system's service life. By more effectively utilizing available inspection information, the PMDP-PH framework supports predictive maintenance within the setting of partially observable Markov deterioration processes. Nonetheless, applying this method in practice is constrained by two main factors. The first relates to data availability, especially information describing system deterioration. In this study, the integration of physical knowledge helps to mitigate this issue. The second limitation concerns the resources required to carry out inspection and maintenance activities. Future work will examine how limited resources influence the feasibility of real-world applications.

6. Conclusion

This paper introduces an innovative methodology that merges PINNs with a PMDP-PH model to support efficient and resilient maintenance planning for infrastructure susceptible to carbonation. By incorporating fundamental physical principles into neural architectures, PINNs enable reliable inference of carbonation behavior, even when observational data is partially informative. The proposed model accounts for general, non-exponential sojourn time distributions during both the initiation and propagation phases of deterioration, which are approximated using hypo-exponential representations. The PMDP-PH framework facilitates the continuous adjustment of maintenance and inspection policies by updating the RUL distribution in real time. This adaptive mechanism is formulated as a multi-stage MPC problem over chosen belief states. This special multi-stage MPC is characterized by contraction mapping and a non-decreasing value function. When applied to a typical infrastructure system, the proposed approach results in a 61.9% reduction in overall maintenance expenditures relative to benchmark strategies, even under variable deterioration scenarios. These findings demonstrate the value of integrating physics-informed learning with a new form of predictive control strategy to bolster infrastructure resilience against increasing carbonation threats.

When applying our designed approach to multi-component systems, it is necessary to consider their internal interdependencies. Moreover, as the number of system components increases, the states, observations, and actions will rise accordingly. In these settings, multi-agent reinforcement learning algorithms will be designed to improve the computational efficiency in future research. We utilize the PINN to capture the deterioration by embedding the physics knowledge into the formulation. Our approach is adaptive to the changes in parameters caused by the types of concrete and environmental conditions. In the future, we will conduct further in-depth research on the changes in concrete types and environmental conditions, mainly focusing on our approach's adaptability.

CRediT authorship contribution statement

Chunhui Guo: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhenglin Liang:** Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. The proof of Proposition 1

Proof. Since $\mathbf{B}(t_d)$ is discrete when representative beliefs are selected, $\mathcal{H}$ maps $\mathbf{V}$ into $\mathbf{V}$, Let u and v be elements in $\mathbf{V}$, and supposed that $\mathcal{H}v(\mathbf{B}(t_d)) \geq \mathcal{H}u(\mathbf{B}(t_d))$. Let a_l^* represent the optimal action at t_d. Then

$$
\begin{aligned}
0 \leq \mathcal{H}v(\mathbf{B}(t_d)) - \mathcal{H}u(\mathbf{B}(t_d)) &\leq \sum_{s_{ki}} \mathbf{B}\left(S_{t_d} = S_{ki}\right) \gamma \sum_{S_{k'i'}} P\left(S_{t_d+\Delta} = S_{k'i'} | S_{t_d}, a_l^*\right) \\
&\sum_{O_{t_d+\Delta}} E\left(O_{t_d+\Delta} | S_{t_d+\Delta}\right) \left[v(\mathbf{B}(t_d+\Delta)) - u(\mathbf{B}(t_d+\Delta))\right] \\
&\leq \gamma \parallel v(\mathbf{B}(t_d+\Delta)) - u(\mathbf{B}(t_d+\Delta)) \parallel
\end{aligned}
\tag{A.1}
$$

By repeating this argument, the condition $\mathcal{H}v(\mathbf{B}(t_d)) \leq \mathcal{H}u(\mathbf{B}(t_d))$ leads to the conclusion that

$$
| \mathcal{H}v(\mathbf{B}(t_d)) - u(\mathbf{B}(t_d)) | \leq \gamma \parallel v(\mathbf{B}(t_d+\Delta)) - u(\mathbf{B}(t_d+\Delta)) \parallel \tag{A.2}
$$

When $T \to \infty$, $V(\mathbf{B})$ converges to a point. □

Appendix B. The proof of Proposition 2

Proof. Proposition 2 is carried out via induction. We begin by assuming that the value function for the PMDP-PH takes the form given in Eq. (19) at the initial time T. Then, we hypothesize that the value function at the decision epoch $t_d + \Delta$ follows the structure presented in Eq. (B.1).

$$
\begin{aligned}
&V(\mathbf{B}(t_d+\Delta)) = \\
&\min_{\mathcal{L}} \frac{\sum_{S_{k'i'}} \alpha^{\mathcal{L}}(S_{t_d+\Delta}) \sum_{s_{ki}} \mathbf{B}(S_{t_d} = S_{ki}) P(S_{t_d+\Delta} = S_{k'i'} | S_{t_d}, a_l) E(O_{t_d+\Delta} | S_{t_d+\Delta})}{\sum_{s_{ki}} \sum_{S_{k'i'}} \mathbf{B}(S_{t_d} = S_{ki}) P(S_{t_d+\Delta} = S_{k'i'} | S_{t_d}, a_l) E(O_{t_d+\Delta} | S_{t_d+\Delta})}
\end{aligned}
\tag{B.1}
$$

The value function at t_d can then be expressed as Eq. (B.2).

$$V\left(\mathbf{B}\left(t_d\right)\right)=\min_{a_l}\left[\sum_{s_{ki}}\mathbf{B}\left(S_{t_d}=S_{ki}\right)\left[q_{S_{ki}}^{a_l}+\sum_{S_{k'i'},O_{t_d+\Delta}}P\left(S_{t_d+\Delta}=S_{k'i'}|S_{t_d},a_l\right)E\left(O_{t_d+\Delta}|S_{t_d+\Delta}\right)\alpha^{\mathcal{L}}\left(S_{t_d+\Delta}\right)\right]\right]\quad\square \tag{B.2}$$

Appendix C. The proof of Proposition 3

Proof. We are tasked with determining whether to implement action a_1 or a_2. When a_2 is selected, the value V_{a_2} remains fixed. To evaluate the value of a_1, we consider two limit scenarios. Scenario 1: If the system operates perfectly, the value $V_{a_1}^1$ is calculated as Eq. (C.1).

$$c_I+\sum_{S_0}\mathbf{B}\left(S_0\right)\sum_{S_{k'i'}}P\left(S_\Delta=S_{k'i'}|S_0,a_1\right)\sum_{O_\Delta}E\left(O_\Delta|S_\Delta\right)V\left(\mathbf{B}\left(\Delta\right)\right) \tag{C.1}$$

Scenario 2: If the system fails, the value $V_{a_1}^2$ is computed as Eq. (C.2).

$$c_I+\sum_{S_{ki}}\mathbf{B}\left(S_{t_d}=S_{ki}\right)\sum_{S_{31}}P\left(S_{t_d+\Delta}=S_{31}|S_{t_d},a_1\right)\sum_{O_{31}}E\left(O_{31}|S_{31}\right)V\left(\mathbf{B}\left(t_d+\Delta\right)\right) \tag{C.2}$$

In practice, the following inequality often holds: $V_{a_1}^1\le V_{a_2}\le V_{a_1}^2$. Next, we demonstrate the monotonicity of $V\left(\mathbf{B}\left(t_d\right)\right)$ under action a_1. We apply induction to show that if $\mathbf{B}\left(t_{d_1}\right)$ is stochastic increasing than $\mathbf{B}\left(t_{d_2}\right)$ under action a_1, the corresponding value functions also exhibit monotonic behavior. We initialize with $V\left([0,\dots,1]\right)=c_4$, where c_4 is a large constant. By this assumption, it follows that: $V\left(\mathbf{B}\left(t_{d_1}\right)\right)-V\left(\mathbf{B}\left(t_{d_2}\right)\right)\ge 0$. Thus, the value function $V\left(\mathbf{B}\left(t_d\right)\right)$ is monotonically increasing. Moreover, there exists a control limit for the optimal selection of action a_2. Finally, when the interval Δ is non-equidistant, Proposition 3 still holds. $\square$

Appendix D. The proof of Proposition 4

Proof. The multi-stage MPC can be represented using a scenario tree. In this framework, the scenario tree assumes a discrete-time formulation for an uncertain system, which can be expressed as:

$$x_{k+1}^c=f\left(x_k^j,u_k^j,d_k^r\right) \tag{D.1}$$

where each state x_{k+1}^c at stage $k+1$ depends on the parent state x_k^j, the corresponding control input u_k^j, and the realization r of the uncertainty at stage k, denoted by d_k^r. The scenario tree is represented by I, The multi-stage MPC problem can then be formulated as

$$J=\min_{x_k^j,u_k^j,\forall(j,k)\in I}\sum_{j\in S}w^j\left(v\left(x_N^j\right)+\sum_{k=0}^{N-1}l\left(x_k^j,u_k^j\right)\right) \tag{D.2}$$

subject to:

$$x_{k+1}^c=f\left(x_k^j,u_k^j,d_k^r\right),\forall(j,k)\in I\backslash I_{leaf} \tag{D.3a}$$

$$x_k^j\in\mathcal{X},\forall(j,k)\in I \tag{D.3b}$$

$$u_k^j\in\mathcal{U},\forall(j,k)\in I\backslash I_{leaf} \tag{D.3c}$$

$$x_N^j\in\mathcal{X}_f,\forall(j,N)\in I \tag{D.3d}$$

where $l\left(x_k^j,u_k^j\right)$ represents the stage cost while $v\left(x_N^j\right)$ denotes the terminal penalty term. The states and control inputs are constrained to lie within the compact sets $\mathcal{X}$, $\mathcal{U}$, with $\mathcal{X}_f$ representing the terminal set. Additionally, the variable w^j denotes the accumulated weight associated with the node x_k^j.

Eq. (16) defines the set of all beliefs that can be reached in a single step. By considering all possible one-step action-observation sequences from these beliefs, we can determine the beliefs that are reachable from $\mathbf{B}(0)$ in two steps, and so on. This process generates a tree structure with the initial belief state $\mathbf{B}(0)$ as the root. The space of reachable beliefs corresponds to all the nodes within this tree. In line with the formulation of multi-stage MPC, the PMDP-PH can thus be reformulated as:

$$Eq.\ (D.2)\rightarrow Eq.\ (17) \tag{D.4}$$

subject to:

$$Eq.(D.3a)\rightarrow\sum_{S_{ki},S_{k'i'},O_{t_d+\Delta}}\mathbf{B}\left(S_{t_d}=S_{ki}\right)P\left(S_{t_d+\Delta}=S_{k'i'}|S_{t_d},a_l\right)E\left(O_{t_d+\Delta}|S_{t_d+\Delta}\right),\ \forall S_{t_d},S_{t_d+\Delta}\in\mathcal{S},O_{t_d+\Delta}\in\mathcal{O},a_l\in\mathcal{A},\mathbf{B}\left(t_d\right)\in\widetilde{\mathcal{B}},t_d\in[0,T] \tag{D.5a}$$

$$Eq.(D.3b)\rightarrow\mathbf{B}\left(t_d\right)\in\widetilde{\mathcal{B}},\forall t_d\in[0,T] \tag{D.5b}$$

$$Eq.(D.3c)\rightarrow a_l\in\mathcal{A} \tag{D.5c}$$

$$Eq.(D.3d)\rightarrow\mathbf{B}(T)\in\widetilde{\mathcal{B}} \tag{D.5d}$$

The state x, control u, and uncertainty d can be reformed as chosen belief state $\mathbf{B}$, action a, and transition probability P and emission E. Eq. (D.2) can be regarded as Eq. (17). $v\left(x_N^j\right)$ can be transformed as $\mathbf{B}(T)\,\mathbf{q}(T)$. $l\left(x_k^j,u_k^j\right)$ is equivalent to the cost of action. Hence, the optimization of the PMDP-PH with chosen belief states can be equivalent to solving the problem of multi-stage MPC. $\square$

Data availability

Data will be made available on request.

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